

Fleet Performance & Delivery Efficiency Dashboard

Power BI Module-End Project – One-Page Summary

Project Overview

Built an end-to-end Power BI dashboard analyzing fleet performance and delivery efficiency for a logistics operation, using trip-level data (50 trips) and vehicle master data (7 vehicles). The project covered data cleaning, data modeling, DAX measures, standard visualizations, and Power BI's AI-powered visuals (Key Influencers, Decomposition Tree).

What Was Done

- Data cleaning: imputed missing Fuel_Consumed values using vehicle-grouped mean imputation (both Excel AVERAGEIF and a DAX calculated column approach).
- Data modeling: created a one-to-many relationship between TripData and Vehicle Master on Vehicle_ID.
- DAX measures: Fuel Efficiency, On-Time Delivery %, Cost per km, Average Cost per km, Average Maintenance Cost, Total Trips.
- Visuals: bar chart (On-Time Delivery % by Destination), line chart (Fuel Efficiency trend by Delivery Date), KPI cards, pie chart (Vehicle Type vs Average Maintenance Cost).
- AI visuals: Key Influencers and Decomposition Tree configured per the assignment brief; Decomposition Tree used to break down Cost per km by Vehicle Type, Maintenance Cost, and Distance.

Key Findings

- On-time delivery performance varies widely by destination, from 87.5% (Delhi, Pune) down to 33.3% (Mumbai).
- Truck is the highest-cost vehicle type per km (~\$18.61), driven primarily by maintenance cost and distance in the Decomposition Tree breakdown.
- Average cost per km across the fleet is \$26.53; average maintenance cost is highest for Trucks (45.25% of total maintenance spend).

Issues Encountered & How They Were Handled

Issue	Details	Resolution / Note
Key Influencers – no results	Configured Analyze = On-Time Delivery % (numeric measure, since the data model requires explicit measures rather than raw text columns) with Distance_km, Vehicle_Type, and Driver_ID as Explain by fields. The visual consistently returned “No influencers found” across multiple field combinations and both Increase/Decrease directions.	Likely due to the small sample size (50 trips) and an already high baseline on-time rate. Documented as a genuine data limitation rather than a configuration error.
Avg. Delivery Time	The assignment brief calls for an “Avg. Delivery Time” card, but the dataset contains only a single Delivery_Date field with no corresponding trip start/ship date, so no duration can be calculated.	Replaced with a Total Trips card (COUNTROWS measure) as a meaningful summary KPI instead.
Q&A visual unavailable	The classic Q&A visual is not available in this Power BI Desktop version/build.	AI-visuals requirement fulfilled instead via Key Influencers and Decomposition Tree; noted in the dashboard and this report for transparency.

Tools & Techniques

- Excel: AVERAGEIF-based mean imputation, data cleaning.
- Power BI: Power Query, data modelling, DAX (DIVIDE, SUMX, AVERAGEX, RELATED, COUNTROWS, SELECTEDVALUE), Key Influencers, Decomposition Tree.